

Aditya Rao

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EDUCATION

McMaster University

Hamilton, ON

B.Eng in Software Engineering (CO-OP)

Sept 2023 – Present

- 4.00/4.00 CGPA - Dean's Honour List
- Relevant Coursework: Digital Systems, OOP, Discrete Math, Software Design, DSA, Computer Architecture

EXPERIENCE

Software Developer Intern

May 2025 – Aug 2026

IBM

Markham, ON

- Built an AI diagnostic platform using IBM Bob to triage Liberty build breaks with 99% accuracy across 100 test failures, reducing incident investigation time by 2 days per break.
- Led modernization of legacy L3 tools (Retain mainframe, EJB/Servlets, Ruby) to React, Spring Boot, Quarkus, and MicroProfile REST microservices, completing a multi-year Salesforce CSP migration.
- Architected an L3 work-distribution analytics app for 400+ developers to track engineering time across CVEs, features, and QA, replacing shared spreadsheets to save 2 days/week of overhead.
- Accelerated CI/CD reliability for WebSphere Application Server by refactoring 20+ legacy FAT test suites into modular JUnit frameworks.
- Diagnosed 50+ multi-platform build failures across Linux, macOS, Windows, and z/OS while maintaining core WebSphere Application Server, WebSphere Liberty, and OpenLiberty runtimes.

Full Stack Developer Intern

May 2024 – Aug 2024

Medium AI

Hamilton, ON

- Developed and maintained React and TypeScript code for Medium AI, contributing to 2 first-place and 1 second-place finishes in the McMaster Startup Survivor competition, and helping secure \$10,000 in funding.
- Engineered real-time doctor-patient voice-based transcription with gRPC and Firebase, achieving sub-500ms streaming latency by load balancing OpenAI Whisper across self-hosted and managed instances.
- Integrated AI diagnostic hooks to analyze transcripts, highlighting clinical insights for attending physicians.

PROJECTS

WebCraft | 🌐 | C++, CMake, Vcpkg, Asynchronous Programming, Github Actions

Apr 2025 – Sept 2025

- Engineered an asynchronous programming library in C++ 23, featuring coroutine-based task management, concurrent streams, and high-performance networking/file I/O modules to simplify cross-platform async development.
- Automated testing and release workflows using gtest (130+ tests), GitHub Actions CI/CD, and published a custom vcpkg port for standardized cross-platform distribution.

Resume Builder | 🌐 | Java, Spring Boot, Next.js, Typescript, Python, FastAPI, LangChain, Docker

Aug 2024 – Feb 2025

- Built an AI-powered resume builder with Next.js, Spring Boot, and FastAPI for real-time editing and version control.
- Integrated LangChain to automate ATS keyword optimization, intelligent scoring, and template tailoring.
- Reduced manual resume tailoring time from ~2 hours to <10 minutes in internal testing

FashioNova | 🌐 🌐 | Typescript, Next.js, Python, Socket.io, OpenCV, Cloudflare AI, Flask

Oct 2024 – Oct 2024

- Built a virtual wardrobe & AI fashion advisor in 36 hours at Hack the Valley 9 (UofT Scarborough).
- Developed an OpenCV, Flask, and Socket.io pipeline for video clothing overlays with Cloudflare AI outfit suggestions.

EXTRA CURRICULARS

Sys-Admin and Fullstack Developer Executive

Nov 2024 – Present

IEEE McMaster Student Branch

Hamilton, ON

- Set up an RPi 3B+ Cluster for a server to handling up to 100 concurrent clients with <50ms response time
- Managed and provisioned a CI/CD pipeline which generates Docker images for our Kubernetes cluster
- Developed IEEE McMaster student branch website using Next.js, leveraging GitHub Projects and Issues as the admin dashboard for managing events and member contributions

TECHNICAL SKILLS

Languages: Java, JavaScript/TypeScript, Python, C#, C++, SQL, C, Verilog

Frameworks: Spring Boot, Jakarta EE, MicroProfile, Hibernate, Express, Next.js, FastAPI, Flask, UWP

Developer Tools: AWS, GCP, Git, Docker, OpenShift, Kubernetes, Infinispan, IBM Bob

Libraries: Pandas, TensorFlow, OpenCV, SciKit Learn, JavaFX, Redux, JQuery, React, MongoDB